

Making the *Model Yours*

Adaptation, creative workflows, and authorship

TONIGHT

Two tracks, one question

TECHNIQUE

How do you teach a model something it has never seen?

Textual Inversion → DreamBooth → Custom Diffusion
→ LoRA → adapters, plus what makes it fast enough
to run on a laptop.

PRACTICE

What happens to artists when models learn styles?

Copyright and living artists' styles, memorized training
images, making a model forget, and whose defaults a
model carries.

Whose model is it?

A foundation model has seen billions of images. You add twenty of yours.
What comes out — whose work is that?

What prompting alone could not fix

1

Drift

Your character's face, costume or prop changes from shot to shot.

2

Sameness

Everything carries the same glossy, default "AI look".

3

Absence

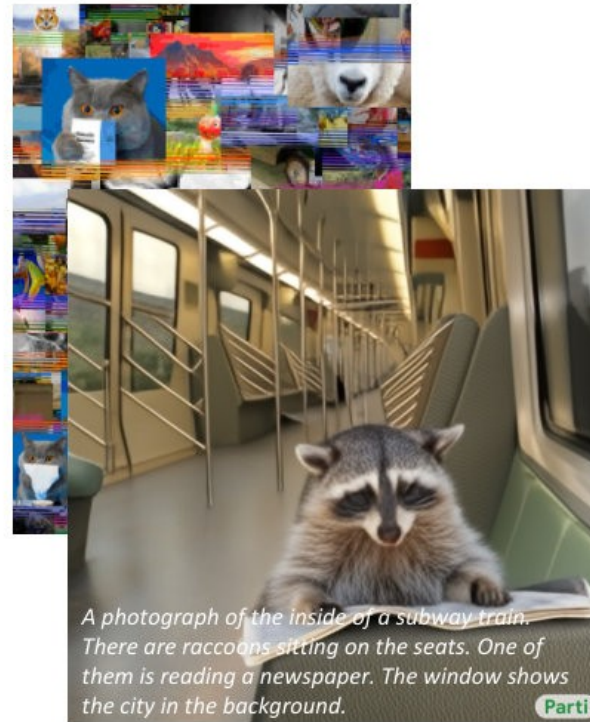
The thing you want — your sketchbook, your city, your grandmother's teapot — is not in the model.

Tonight: moving from *prompting* a model to **shaping** one.

Large-scale Text-to-Image Models



Diffusion models
(DALL-E 2, Imagen, SD)



Autoregressive models
(Image GPT, Parti)



GANs, Masked GIT
(GigaGAN, MUSE)

Limitations of Text-to-Image Models

Linguistic bottleneck: not everything can be described by text

Data bottleneck: many things are not included in the dataset:

1. Not in the public domains (e.g., personal concepts)
2. Have not been created (e.g., new concepts)

Text-to-image isn't perfect...

Stable
Diffusion



Actual **moongate** images

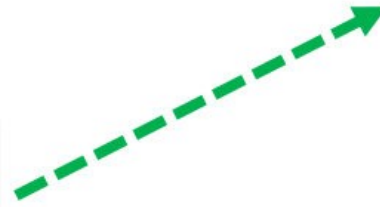


Photo of a **moongate**

Customization



Actual **moongate** images



Stable
Diffusion



Photo of a **moongate**

Customization



Actual **moongate** images

Customized
Diffusion



Photo of a **moongate**

Unseen contexts



Actual **moongate** images

Customized
Diffusion



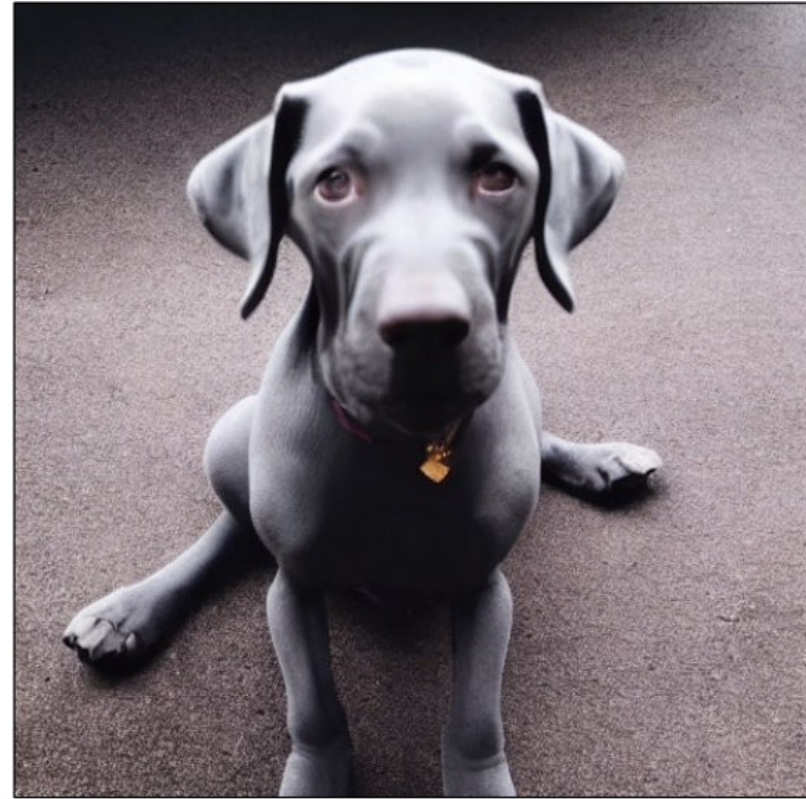
Moongate in snowy ice

No knowledge of personal concepts



My **dog**, Stark

Stable
Diffusion



A dark grey color weimaraner dog

Customization



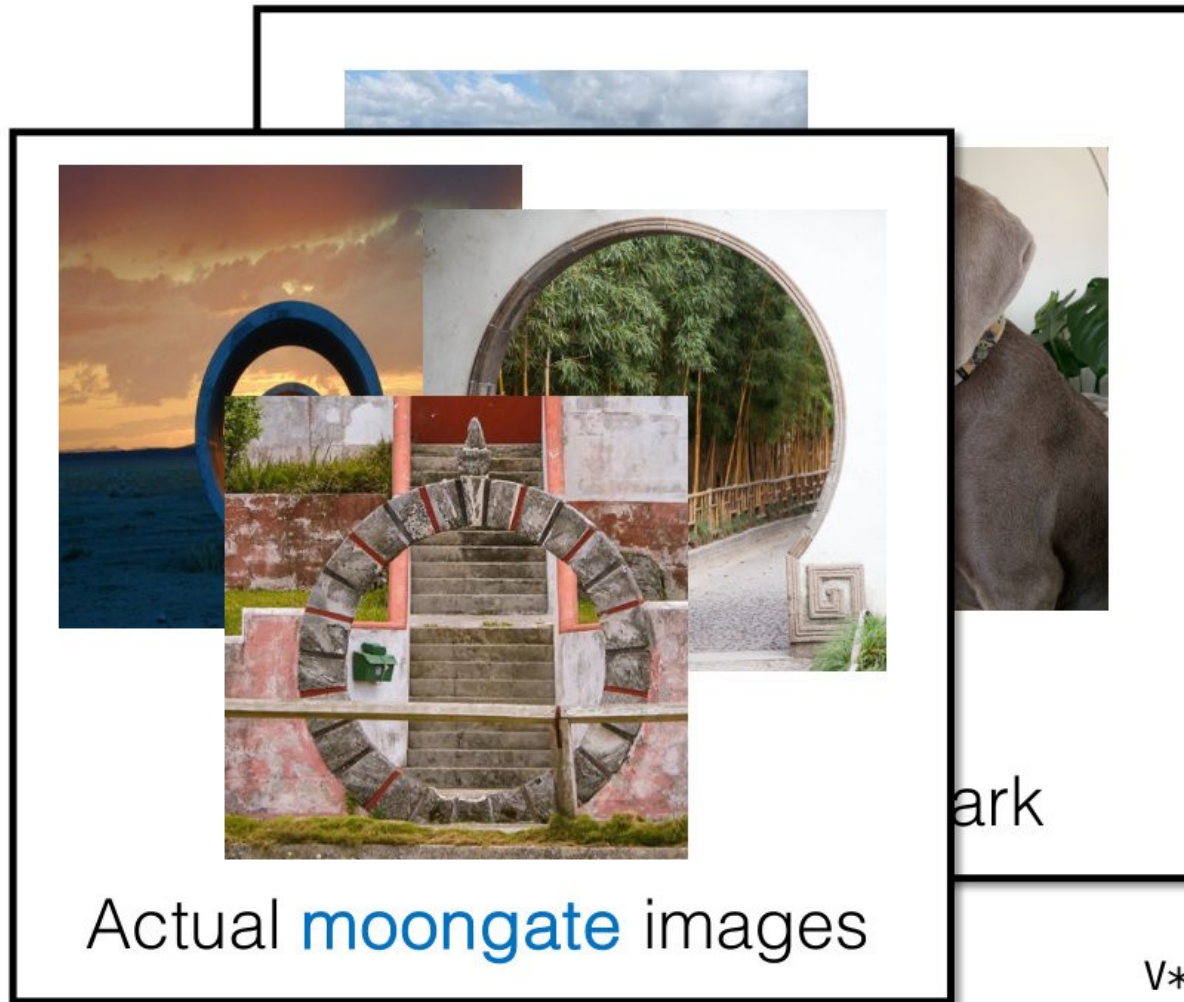
Jun-Yan's **dog**, Stark

Customized
Diffusion



V* **dog** wearing sunglasses

Multiple concepts



Customized
Diffusion



V* **dog** wearing sunglasses in front of **moongate**

Copyrights

- + Disclaimer: I am not a lawyer
 - + Human creator's rights
 - + Diverse opinions
 - + Evolving landscape

Copyrighted content?

- Copyrighted images
- Company IPs / logos
- Artist styles of living artists



Getty Images



Greg Rutkowski

Ongoing Legal Battles

ARTIFICIAL INTELLIGENCE / TECH / LAW

Getty Images sues AI art generator Stable Diffusion in the US for copyright infringement



An illustration from Getty Images' lawsuit, showing an original photograph and a similar image (complete with Getty Images watermark) generated by Stable Diffusion. Image: Getty Images

Getty Images has filed a lawsuit in the US against Stability AI, creators of open-source AI art generator Stable Diffusion, escalating its legal battle against the firm.

/ Getty Images has filed a case against Stability AI, alleging that the company copied 12 million images to train its AI model 'without permission ... or compensation.'

By JAMES VINCENT

Feb 6, 2023, 11:56 AM EST | 16 Comments / 16 New



ARTIFICIAL INTELLIGENCE / TECH / CREATORS

AI art tools Stable Diffusion and Midjourney targeted with copyright lawsuit



A collage of AI-generated images created using Stable Diffusion. Image: [The Verge via Lexica](#)

/ The suit claims generative AI art tools violate copyright law by scraping artists' work from the web without their consent.

By JAMES VINCENT

Jan 16, 2023, 6:28 AM EST | 28 Comments / 28 New



A trio of artists have launched a lawsuit against Stability AI and Midjourney, creators of AI art generators Stable Diffusion and Midjourney, and artist portfolio platform DeviantArt, which recently created its own AI art generator, DreamUp.

Source: The Verge

Ongoing Legal Battles

REUTERS® World Business Markets Legal Breakingviews Technology Investigations Sports

Copyright Technology Intellectual Property Litigation Data Privacy

2 minute read · February 22, 2023 8:41 PM EST · Last Updated 2 months ago

AI-created images lose U.S. copyrights in test for new technology

By Blake Brittain



REUTERS/Andrew Kelly

Feb 22 (Reuters) - Images in a graphic novel that were created using the artificial-intelligence system Midjourney should not have been granted copyright protection, the U.S. Copyright Office said in a letter seen by Reuters.

I'm not so sure. As we've seen, a key assumption for a "non-expressive use" defense is that Stable Diffusion only learns uncopyrightable facts—not creative expression—from its training images. That's *mostly* true. But it's not entirely true. And the exceptions could greatly complicate Stability AI's legal defense.

Stable Diffusion's copying problem

Here's one of the most awkward examples for Stability AI:

Training Set



Caption: Living in the light with Ann Graham Lotz

[Enlarge](#)

Generated Image



Prompt: Ann Graham Lotz

<https://arstechnica.com/>

Which parts shall we customize?

Textual Inversion: Optimizing Text Embedding



[Rinon Gal et al., ICLR 2023]

Textual Inversion Results



[Rinon Gal et al., ICLR 2023]

Works well for artistic styles



Input samples



“The streets of Paris
in the style of S_* ”



“Adorable corgi
in the style of S_* ”



“Painting of a black hole
in the style of S_* ”



“Times square
in the style of S_* ”

[Rinon Gal et al., ICLR 2023]

Cannot preserve object identity



Target images

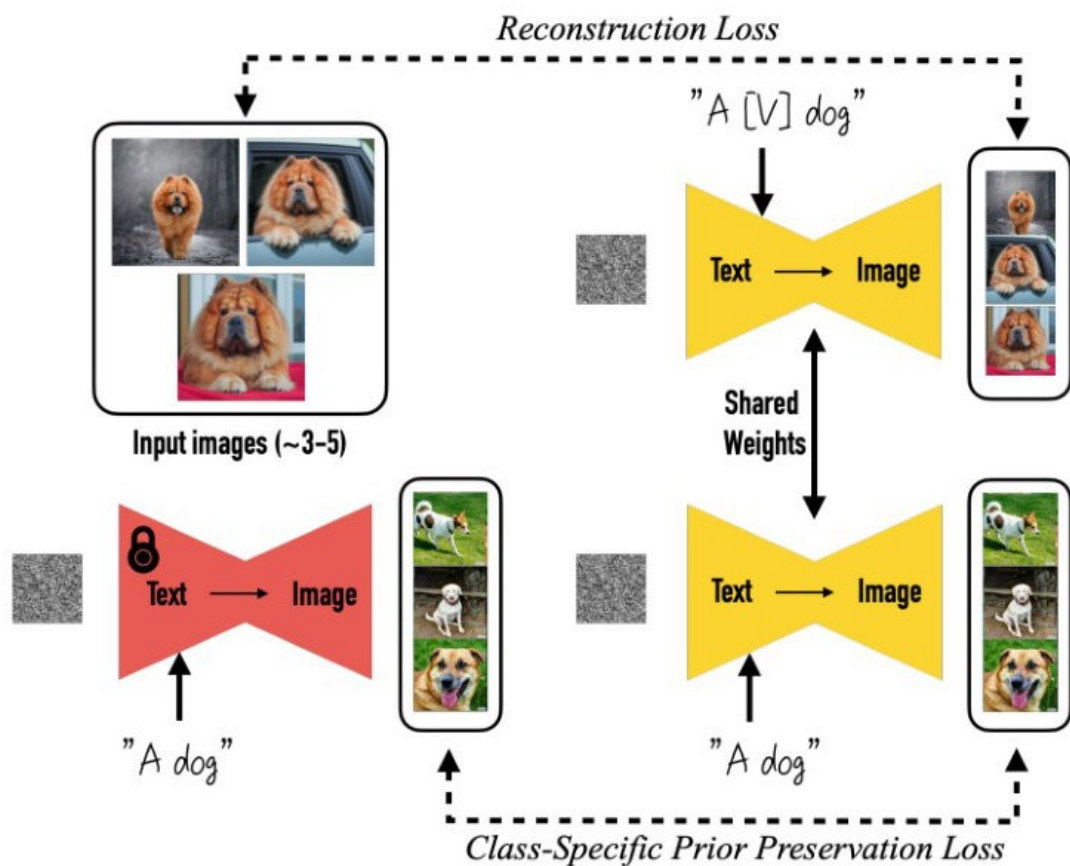


S^* cat swimming in a pool

[Rinon Gal et al., ICLR 2023]

How to improve identity preservation?

DreamBooth: Fine-tuning all the weights



Training Objective

$$\Delta\theta^* = \arg \min_{\Delta\theta} \mathbb{E}_{\epsilon, \mathbf{x}, \mathbf{c}, t} [||\epsilon - \epsilon_{\theta}(\mathbf{x}_t, \mathbf{c}, t)||_2]$$

where $\theta = \theta_0 + \Delta\theta$

Issues (Overfitting)

- Forget to generate subjects of the same class (e.g., dog)
- Reduce output diversity

Regularization

- Add synthetic images of the same class.

Inspired by single-image GAN fine-tuning
GANPaint [Bau et al., 2019], PTI [Roich et al., 2021]

[Nataniel Ruiz et al., CVPR 2023]

DreamBooth Results



Input images



A [V] backpack in the Grand Canyon



A wet [V] backpack in water



A [V] backpack in Boston



A [V] backpack with the night sky



Input images



A [V] teapot floating in milk



A transparent [V] teapot with milk inside



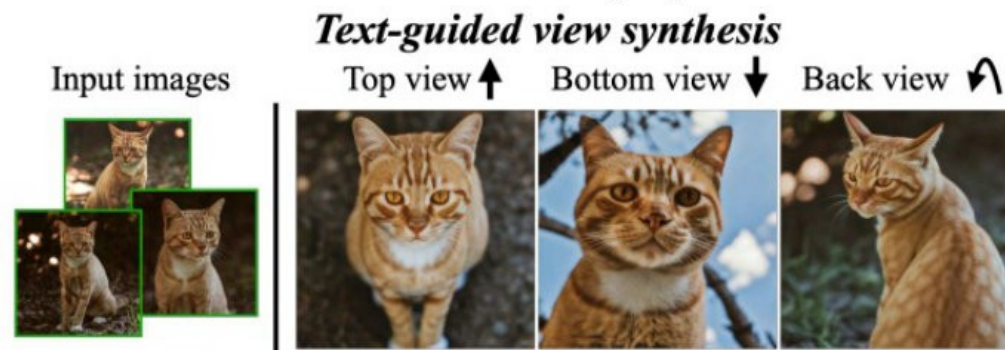
A [V] teapot pouring tea



A [V] teapot floating in the sea

[Nataniel Ruiz et al., CVPR 2023]

DreamBooth Applications



[Nataniel Ruiz et al., CVPR 2023]

DreamBooth vs. Textual Inversion



[Nataniel Ruiz et al., CVPR 2023]

Memorized training images

Real Image



Stable
Diffusion



Ann Graham Lotz

Extracting Training Data from Diffusion Models [Carlini et al., 2023]

Memorized training images

- Step 1: Identifying duplicates in the training data
- Step 2: Generating many images with the selected prompt
- Step 2: Image matching

Original:



Generated:



Extracting Training Data from Diffusion Models [Carlini et al., 2023]

Memorized style

Greg Rutkowski



*image credit: <https://rutkowski.artstation.com>

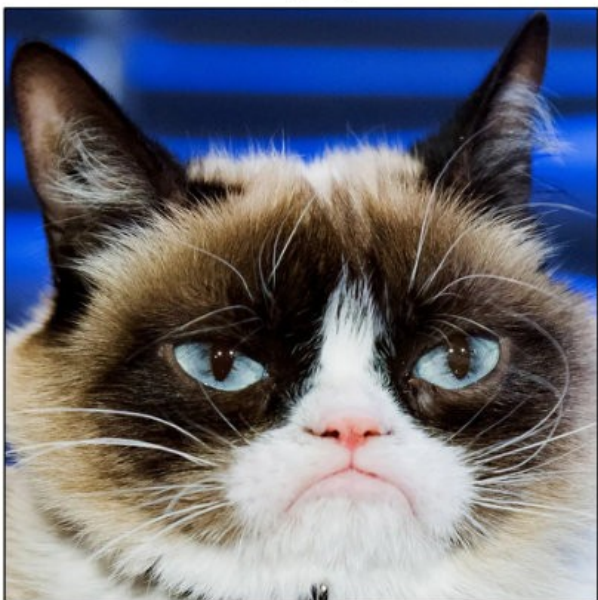
Stable
Diffusion



A painting of a boat on the water
in the style of Greg Rutkowski

Memorized instances

Grumpy cat



Stable
Diffusion



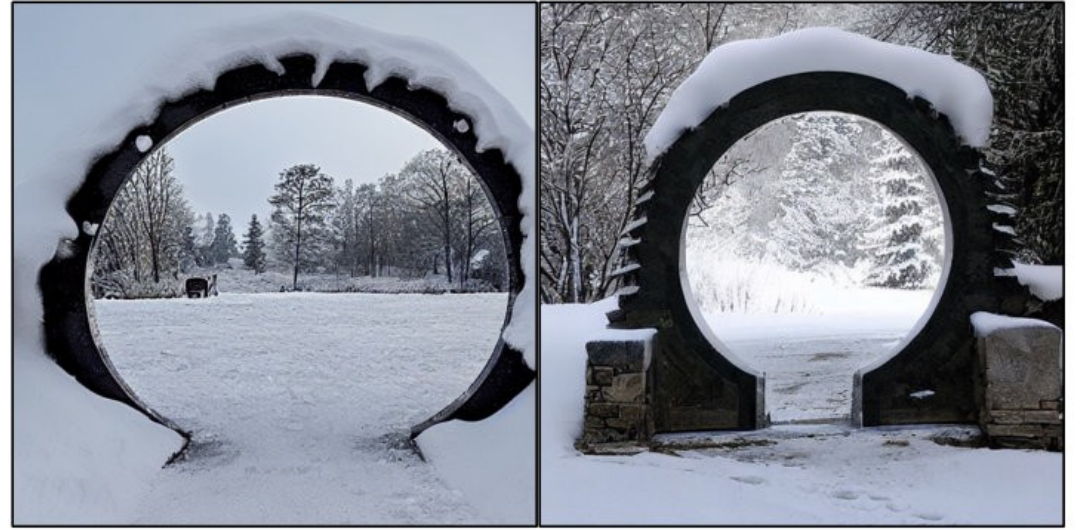
What a cute Grumpy cat

Fine-tuning all model weights

Photo of a **moongate**



Moongate in snowy ice



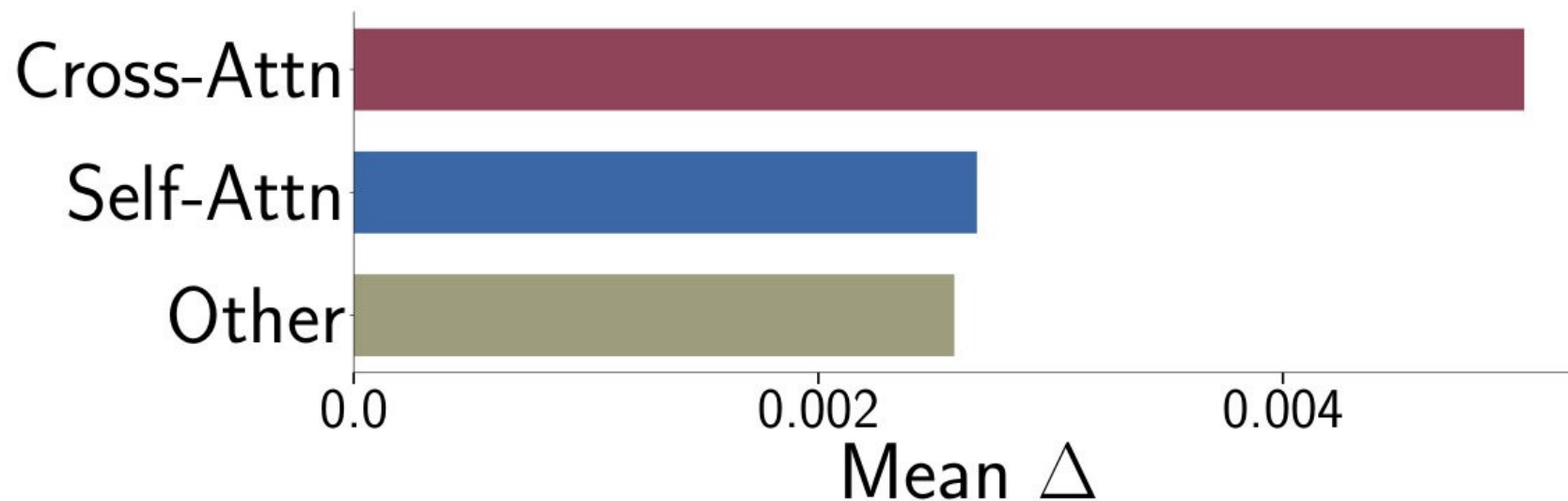
Storage requirement. 4GB storage for each fine-tuned model.

Compute requirement. It requires more VRAM/training time.

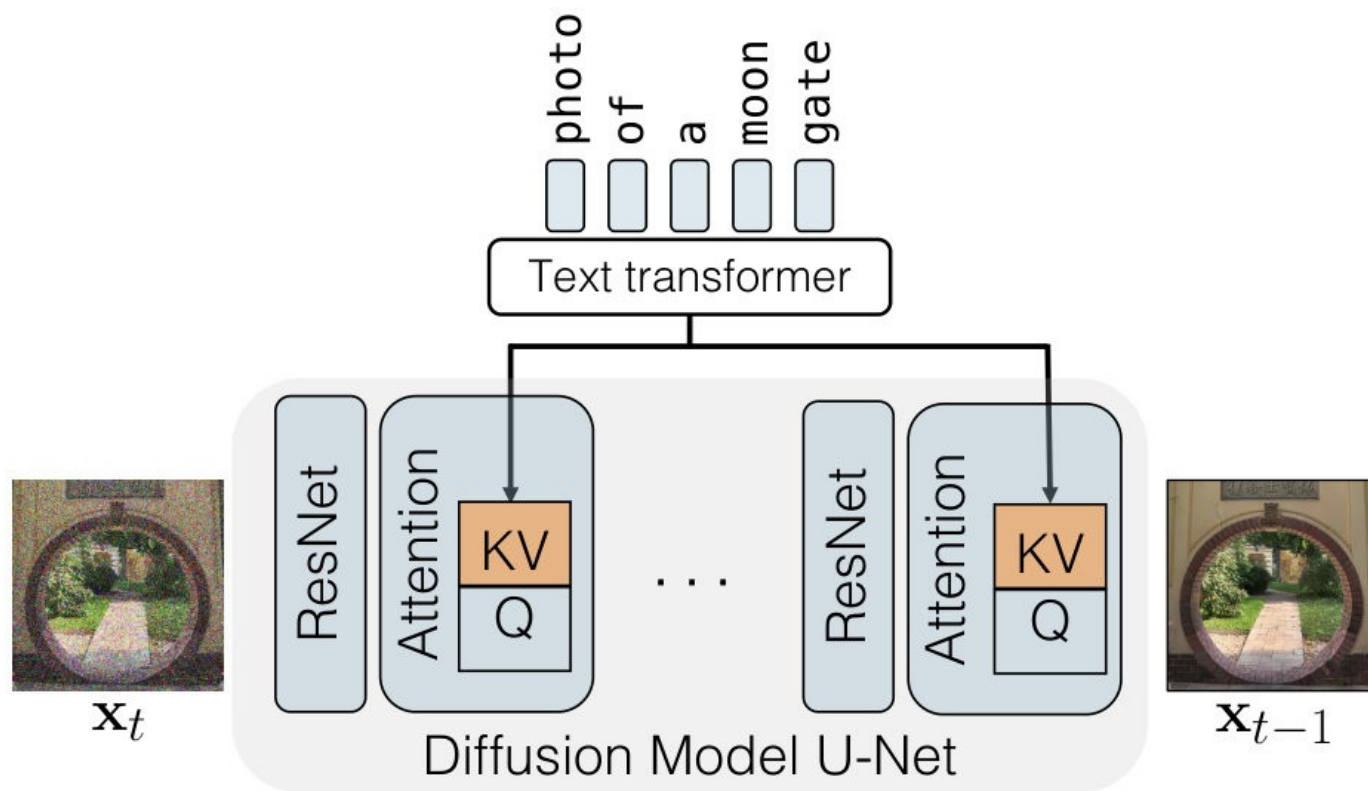
Compositionality. Hard to combine multiple models.

Analyze change in weights

$$\Delta_l = \frac{||\theta'_l - \theta_l||}{||\theta_l||} \quad \text{where} \quad \begin{array}{l} \theta'_l : \text{updated weights} \\ \theta_l : \text{pretrained weights} \end{array}$$



Only fine-tune cross-attention layers



$$\Delta W_k^*, \Delta W_v^* = \arg \min_{\Delta W_k, \Delta W_v} \mathbb{E}_{\epsilon, \mathbf{x}, \mathbf{c}, t} [\|\epsilon - \epsilon_{\theta}(\mathbf{x}_t, \mathbf{c}, t)\|_2]$$

 Trainable  Frozen

[Nupur Kumari et al., CVPR 2023]

How to prevent overfitting?



Photo of a
{moongate}

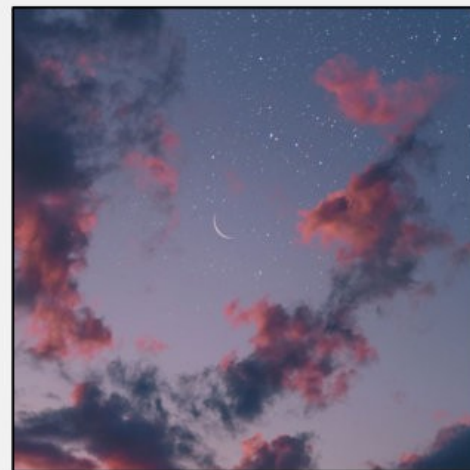


Photo of a
{moongate}

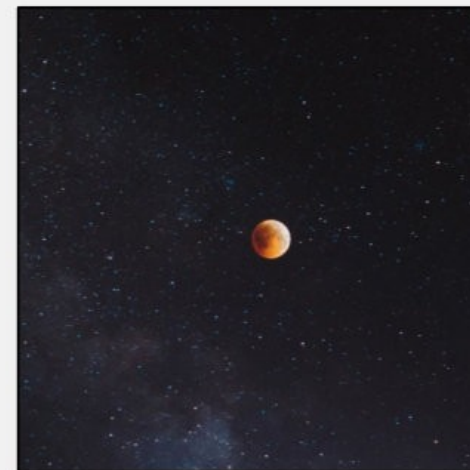
...

Target images

+



sky full of stars
and the moon



Blood moon

...

Add regularization images

Personalized concepts



Jun-Yan's **dog**, Stark

How to describe personalized concepts?

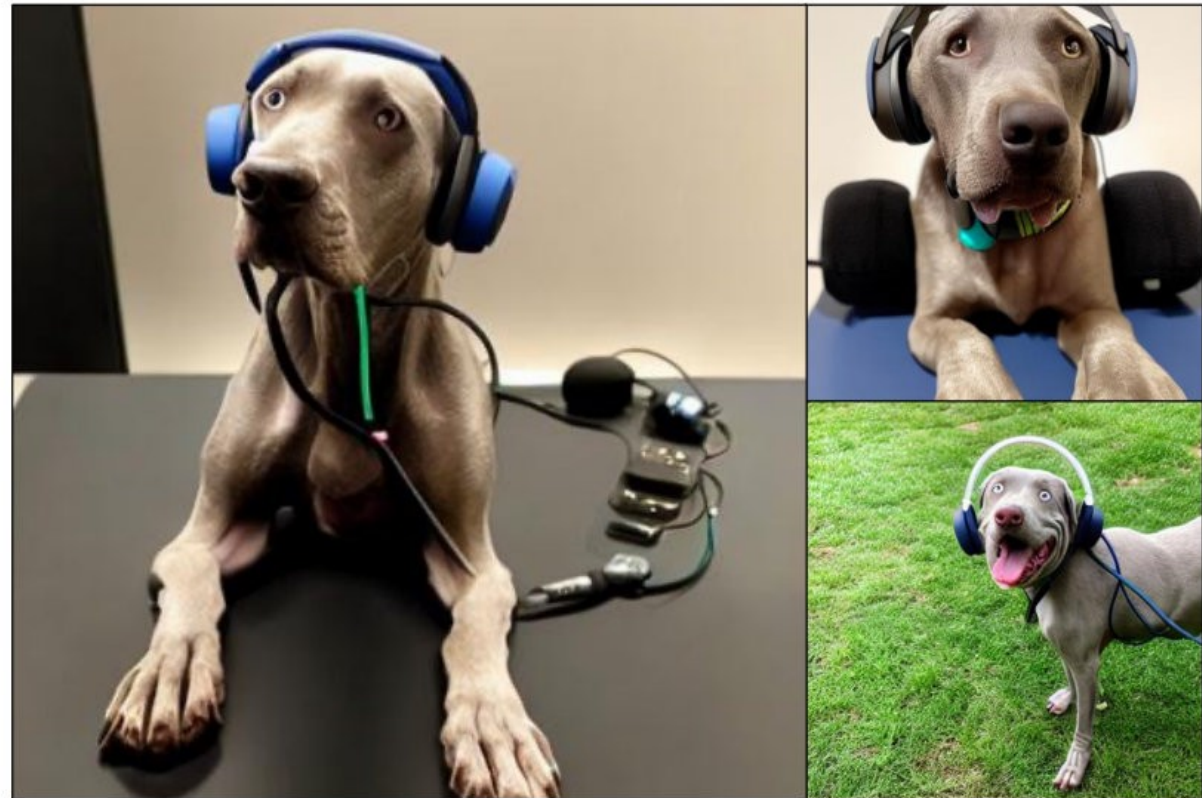
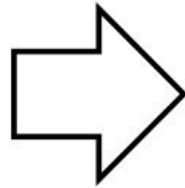
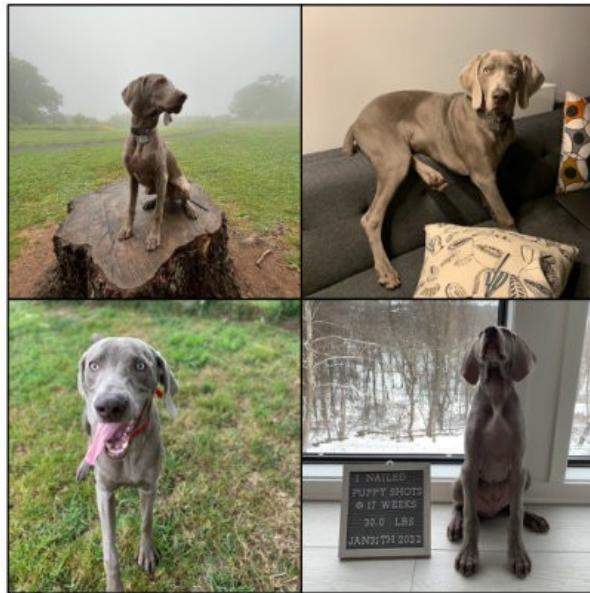
V^* dog

Where V^* is a modifier token in the text embedding space

Proposed by Textual Inversion [Rinon Gal et al.]

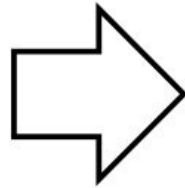
[Nupur Kumari et al., CVPR 2023]

Single concept results



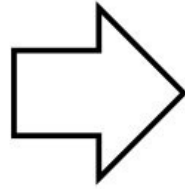
V^* dog wearing headphones

Single concept results



A watercolor painting of V^*
tortoise plushy on a mountain

Results: specific art style

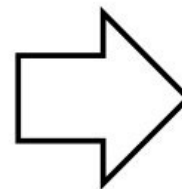


Painting of dog in the style
of V^* art



Drawings from Aaron
Hertzmann

Two concept results



V_1^* art style painting
of V_2^* wooden pot



Drawings from Aaron
Hertzmann

Qualitative comparison (multi-concept)

Target Images



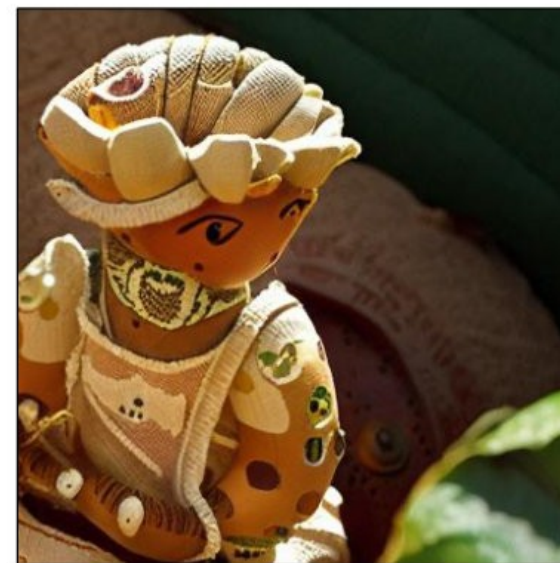
Custom Diffusion (Ours)



DreamBooth



Textual Inversion



V_1^* flower in the V_2^* wooden pot on a table

Limitations



Ours



V_1^* dog and a V_2^* cat
playing together

Pretrained model

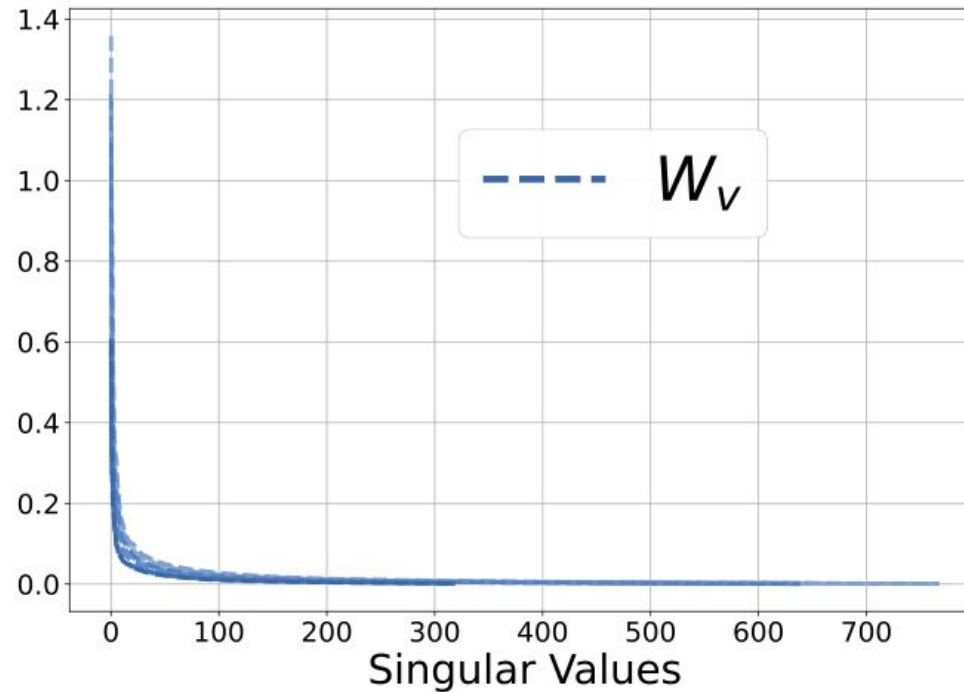


dog and a cat
playing together

Memory requirement

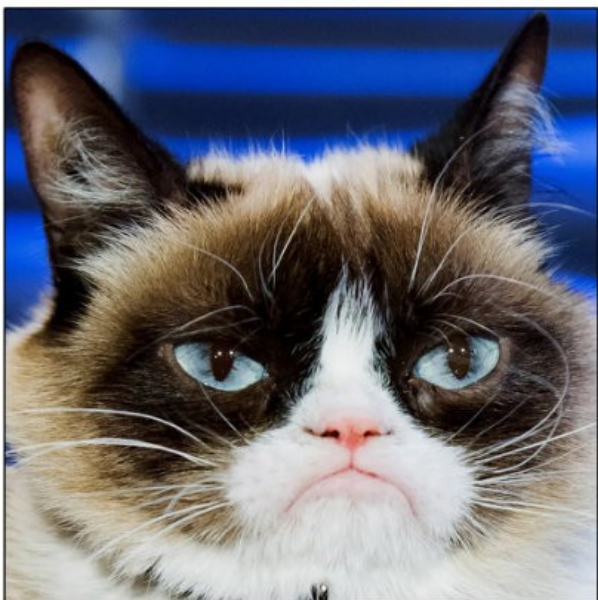
Each custom diffusion model: 75MB storage

Analyze the difference in pretrained and fine-tuned weights

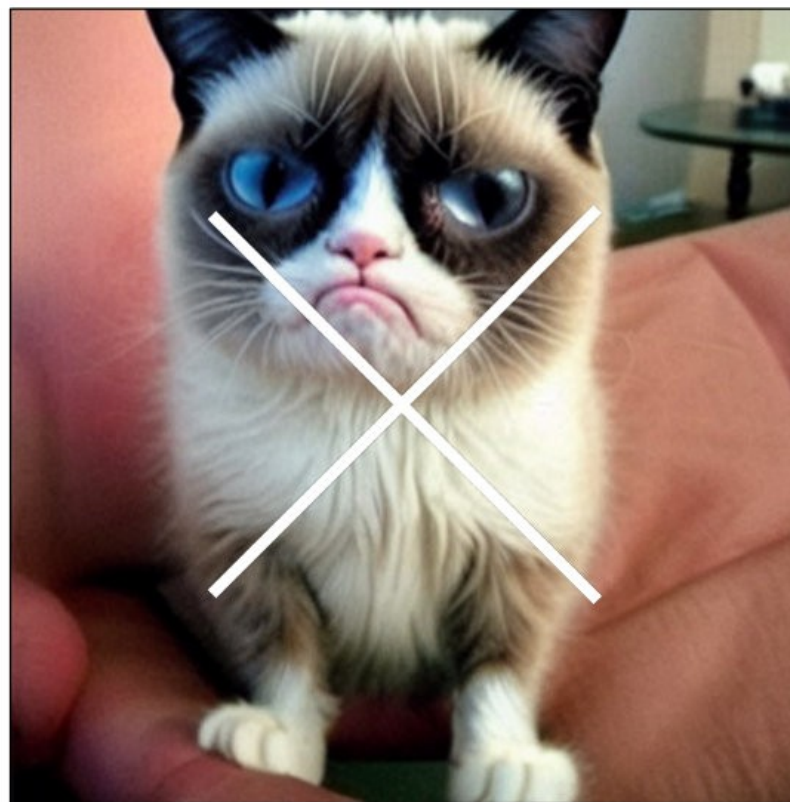


Concept ablation

Grumpy cat



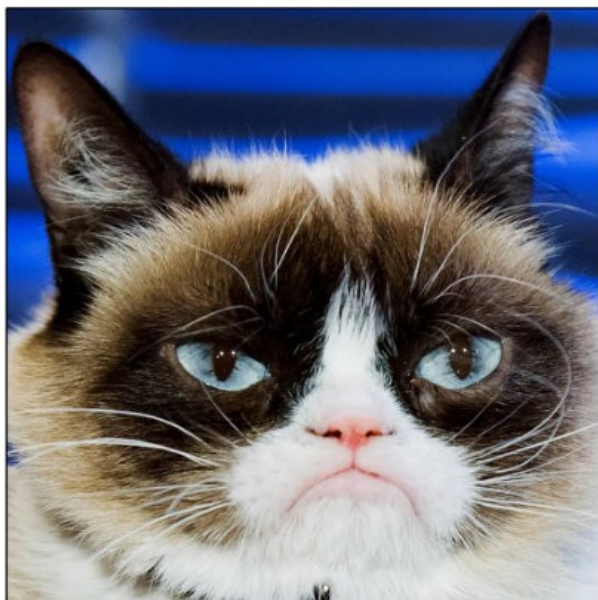
Ablated Stable
Diffusion



What a cute Grumpy cat

Concept ablation

Grumpy cat



Ablated Stable
Diffusion



Generate a random
cat

What a cute Grumpy cat

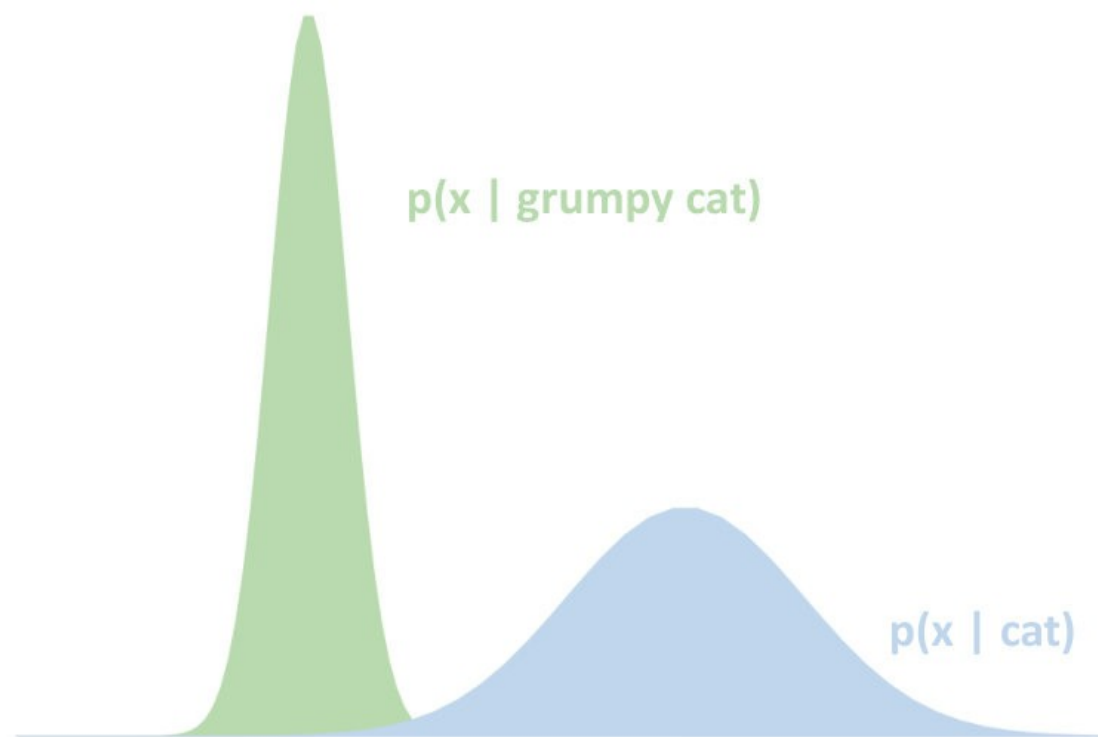
[Kumari et al., arXiv 2023]

Concept ablation



[Kumari et al., arXiv 2023]

Concept ablation



[Kumari et al., arXiv 2023]

Ablating R2D2

Stable
Diffusion



Ablated Stable
Diffusion



The future is now with this amazing home automation R2D2

[Kumari et al., arXiv 2023]

Ablating R2D2

Stable
Diffusion



Ablated Stable
Diffusion



The possibilities are endless with this versatile R2D2

[Kumari et al., arXiv 2023]

Ablating Snoopy

Stable
Diffusion



Ablated Stable
Diffusion



A devoted Snoopy accompanying its owner on a road trip

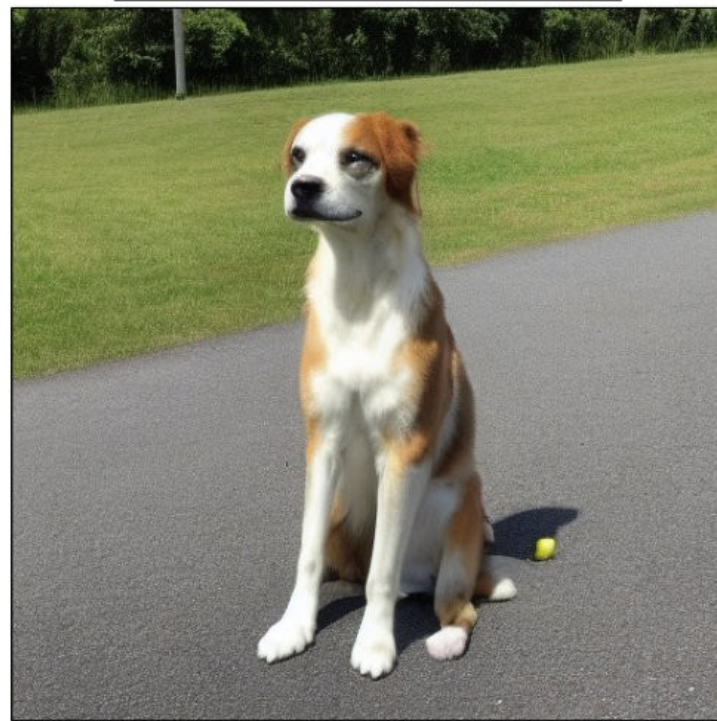
[Kumari et al., arXiv 2023]

Ablating Snoopy

Stable
Diffusion



Ablated Stable
Diffusion



A confident Snoopy standing tall and proud after a
successful training session

[Kumari et al., arXiv 2023]

Ablating Van Gogh



Painting of olive trees in the style of Van Gogh

[Kumari et al., arXiv 2023]

Ablating Van Gogh

Stable
Diffusion



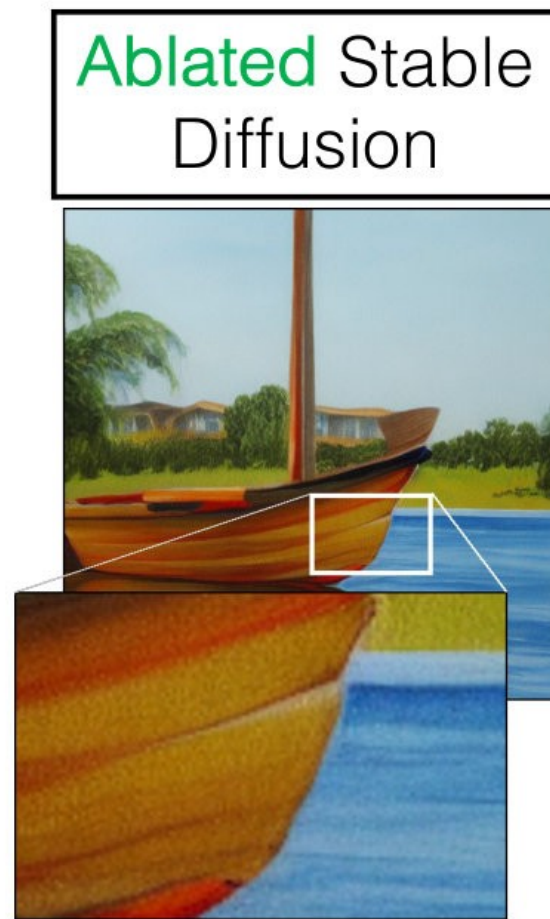
Ablated Stable
Diffusion



Painting of women working in the garden, in the style of
Van Gogh

[Kumari et al., arXiv 2023]

Ablating Greg Rutkowski



A painting of a boat on the water in the style of Greg
Rutkowski

[Kumari et al., arXiv 2023]

Ablating Greg Rutkowski



Painting of a group of people on a dock by Greg Rutkowski

[Kumari et al., arXiv 2023]

Ablating memorized images

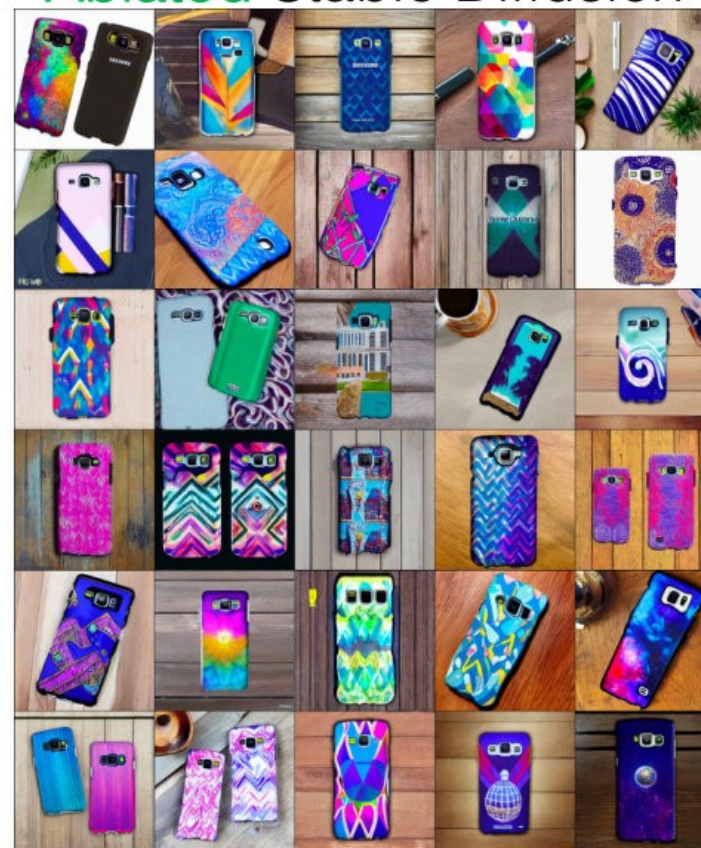
Real Image



Stable Diffusion



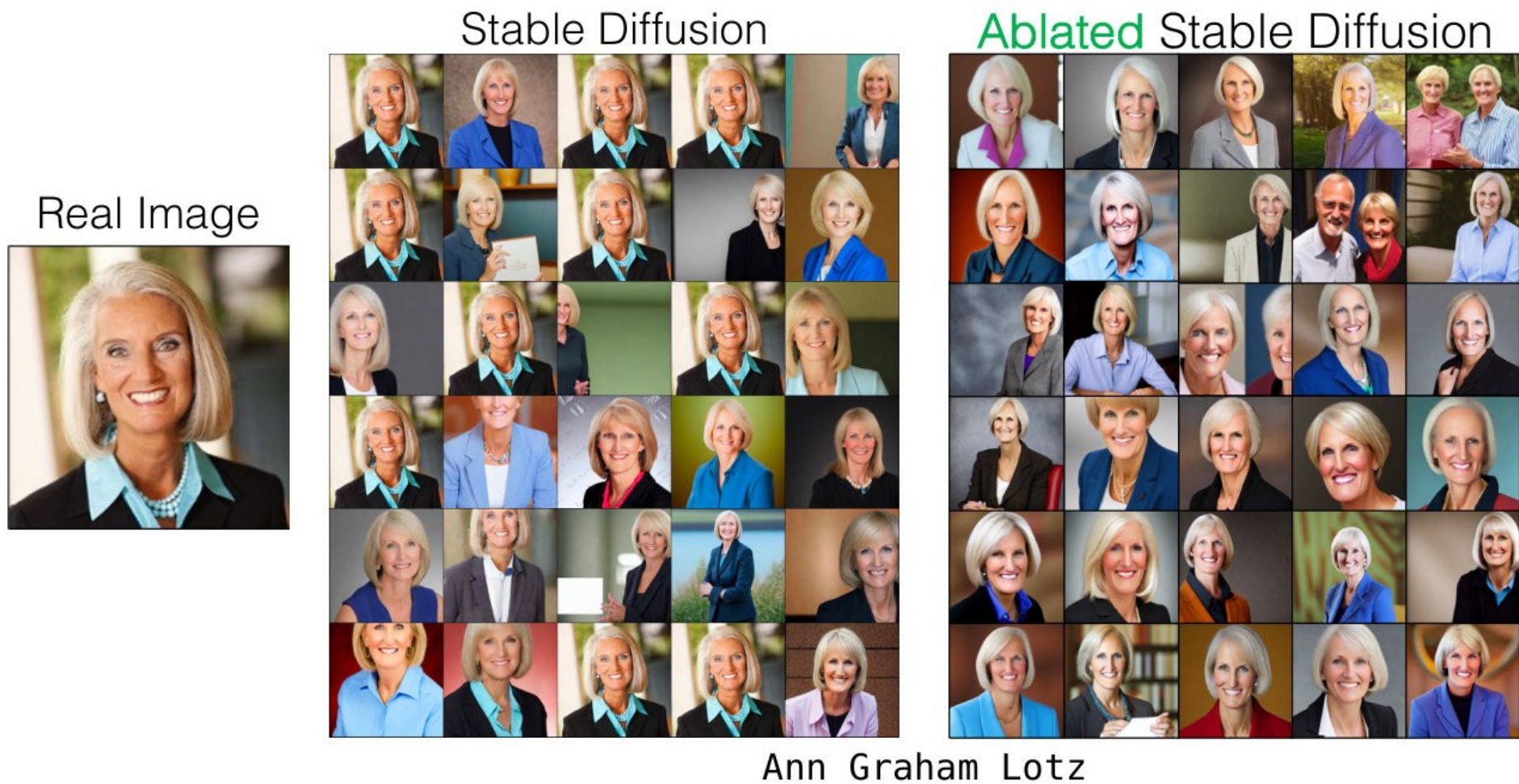
Ablated Stable Diffusion



New Orleans House Galaxy Case

[Kumari et al., arXiv 2023]

Ablating memorized images



[Kumari et al., arXiv 2023]

Ablating composition of concepts

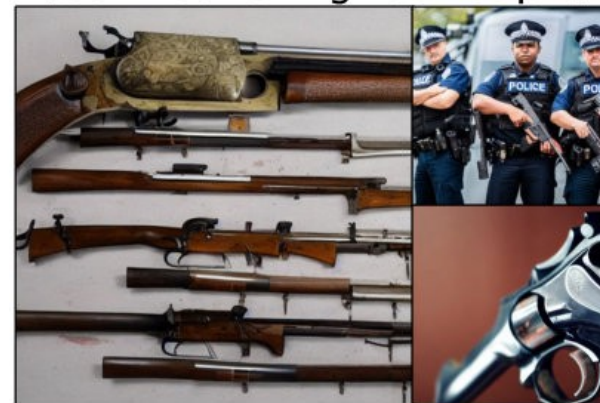
Kids with guns
(target concept)



Kids
(anchor concept)

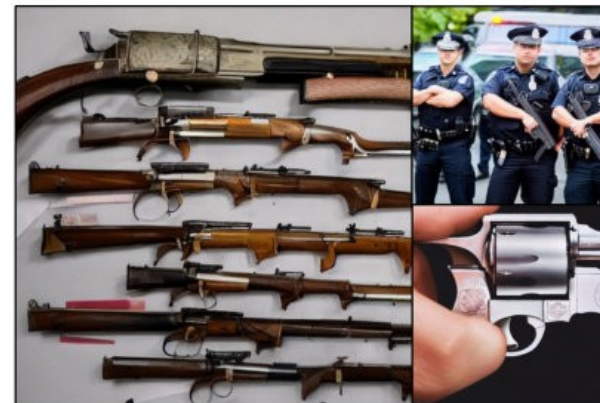


Guns
(surrounding concept)



Stable
Diffusion

Ablated
Stable
Diffusion



[Kumari et al., arXiv 2023]

Other works

1. **Erasing Concepts from Diffusion Models.** Rohit Gandikota, Joanna Materzynska, Jaden Fiotto-Kaufman, and David Bau. arXiv preprint arXiv:2303.07345 (2023).
2. **Forget-Me-Not: Learning to Forget in Text-to-Image Diffusion Models.** Eric Zhang, Kai Wang¹, Xingqian Xu, Zhangyang Wang, Humphrey Shi. arXiv preprint arXiv:2303.17591 (2023).

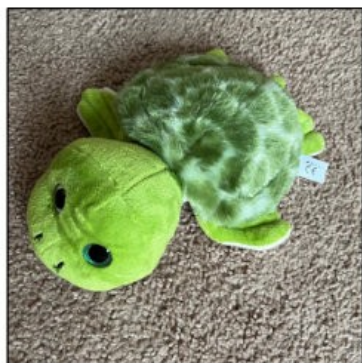
Compressing fine-tuned weights

75MB

15MB

0.1MB

0.08MB



Target image

Custom Diffusion

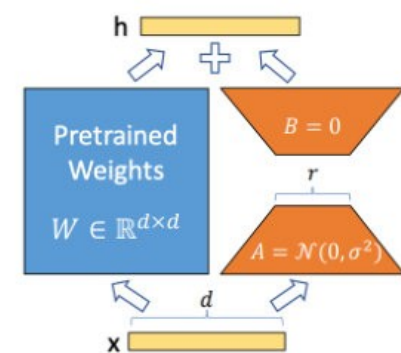
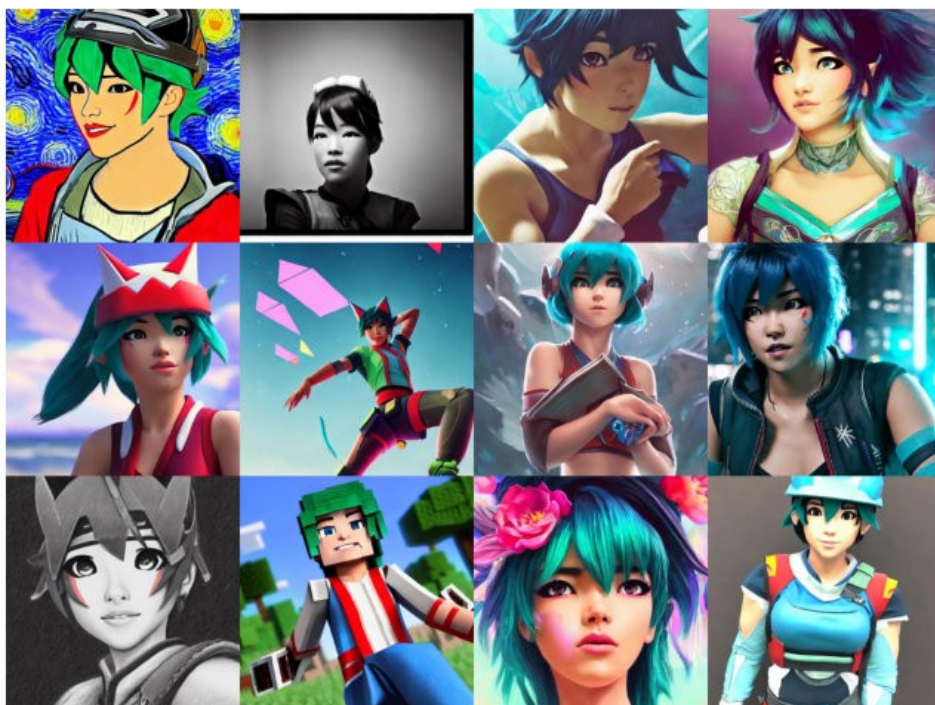
Top 20% rank

1 Rank

0 Rank

Low-rank Adaptation (Lora)

- Lora: Low-rank adaptation of large language models



Original weights

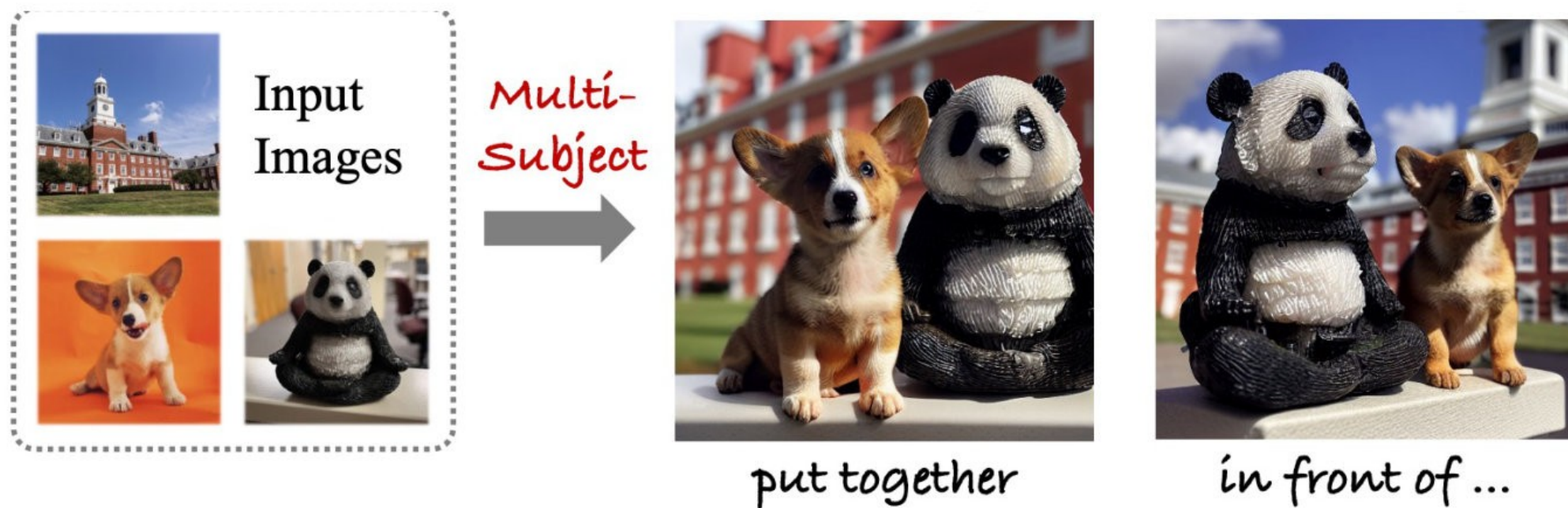
$$W = \overset{\downarrow}{W_0} + BA$$

$\uparrow$
Low-rank difference

Lora [Edward J. Hu*, Yelong Shen*, et al., ICLR 2022]

Lora + Dreambooth (by Simo Ryu): <https://github.com/cloneofsimon/lora>

Low-rank Adaptation (SVDiff)



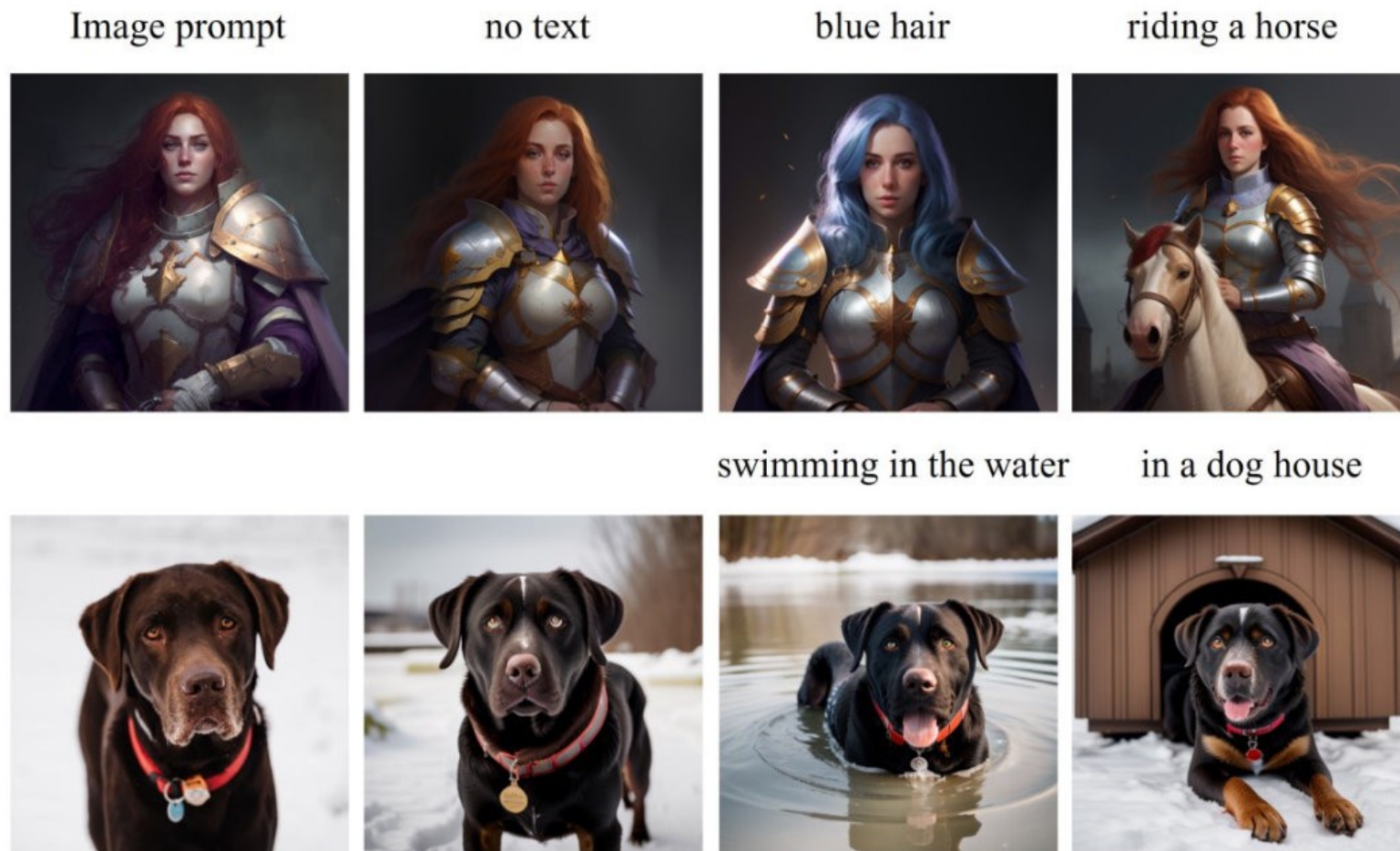
- Composing multiple concepts

$$\Sigma_{\delta'} = \text{diag}(\text{ReLU}(\sigma + \delta_1 + \delta_2)).$$

SVDiff [Han et al., ICLR 2022]

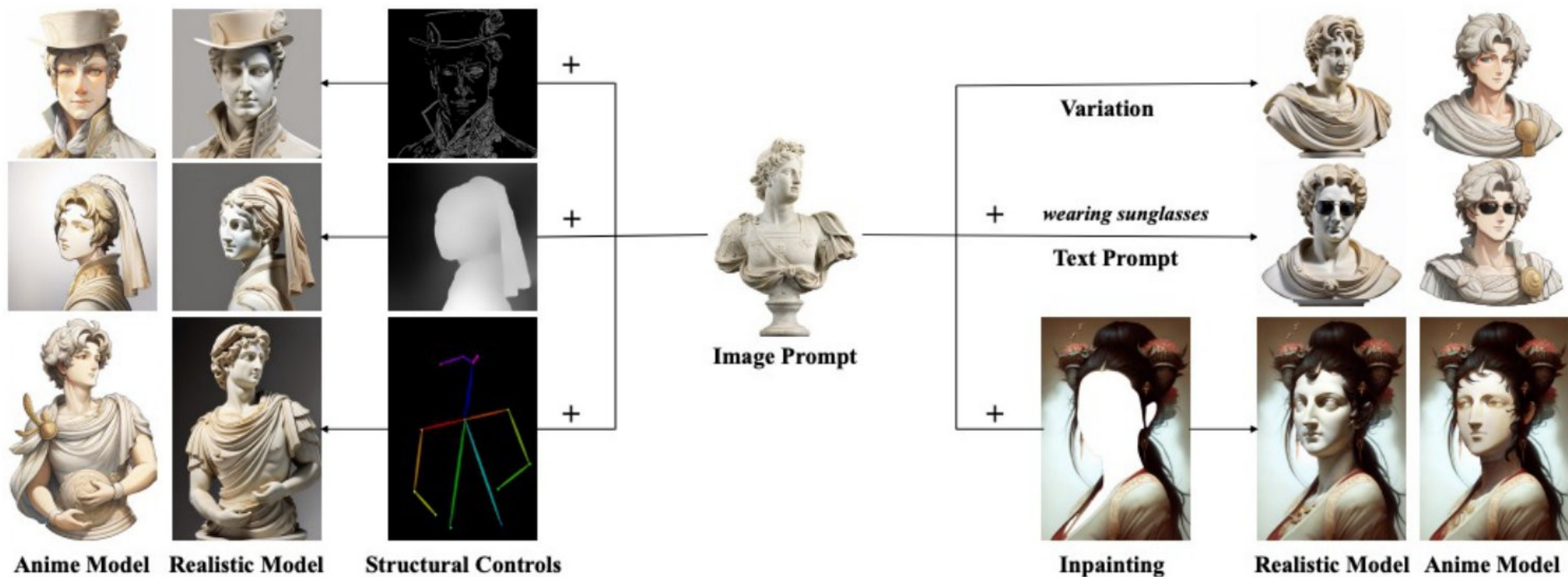
Optimization is too Slow!

Image Prompt Adapter (IP-Adapter)



[He Yu et al., CVPR 2024]

Image Prompt Adapter (IP-Adapter)



[He Yu et al., CVPR 2024]

Optimization + encoder (5-15 steps)



[Rinon Gal et al., arXiv 2023]

Smaller changes, fewer steps, smaller numbers

ADAPTERS

Change less

LoRA trains a thin overlay, not the whole model. The file you share is small.

DISTILLATION

Step less

A "student" model learns to do in a few denoising steps what the "teacher" does in many. Near-real-time generation.

QUANTIZATION

Store less

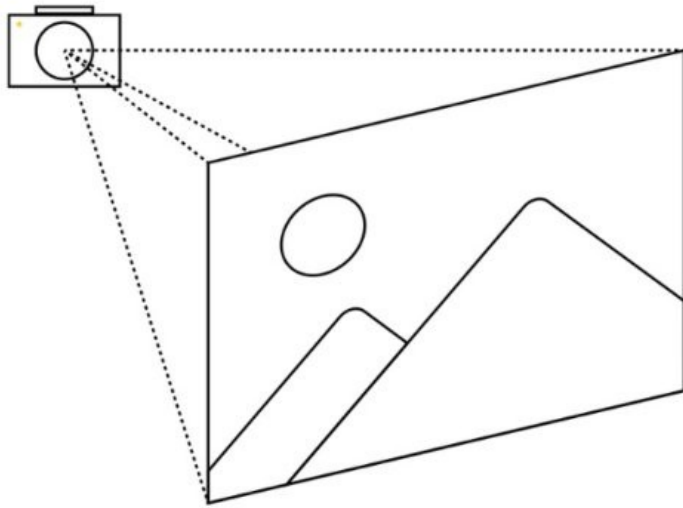
Keep each weight with fewer bits — like saving a photo as a smaller JPEG. It fits on a consumer GPU.

Open or hosted? A creative trade-off

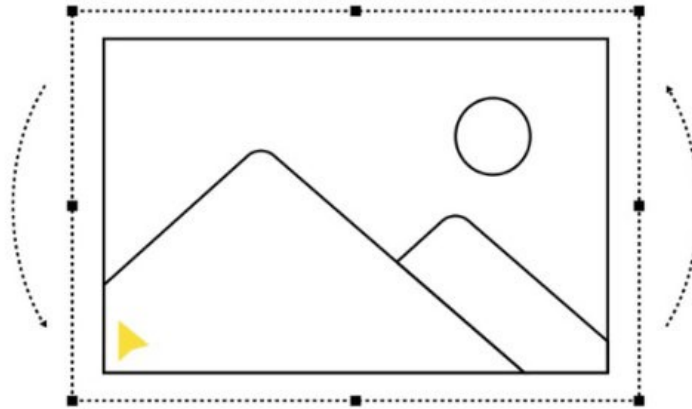
	Open model, local (e.g. ComfyUI)	Hosted tool (web app)
Train your own LoRA	Yes, full control	Sometimes, limited
Where your images go	Stay on your machine	Uploaded to a company
Reproducible	Seed + workflow file	Often not
Effort and cost	Setup, a GPU	Easy, subscription
Frontier quality	Usually a step behind	Often the best available

Your workflow file, seeds and model versions are also your **documentation of AI use** for A2.

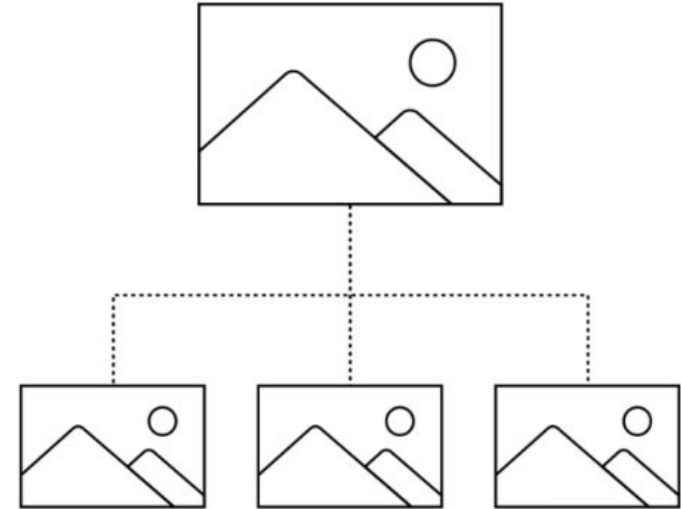
Content Authenticity (prove what is real)



Capture



Editing



Publishing and sharing

<https://contentauthenticity.org/>

Biases

Danger and Ethical Concerns

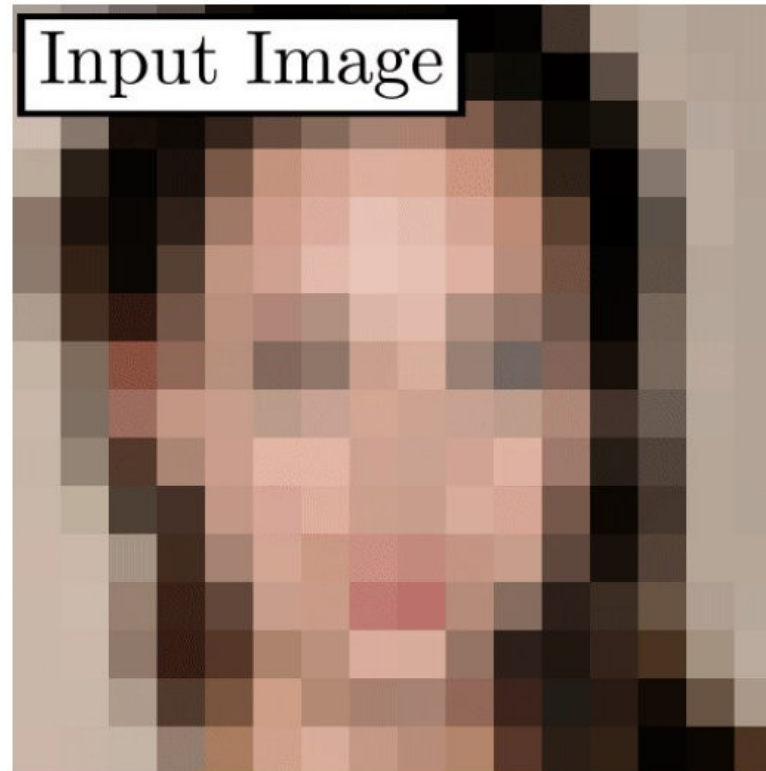


Image Super-resolution system PULSE [Menon et al., CVPR 2020]

Super-resolution with GANs Inversion and StyleGAN

Danger and Ethical Concerns

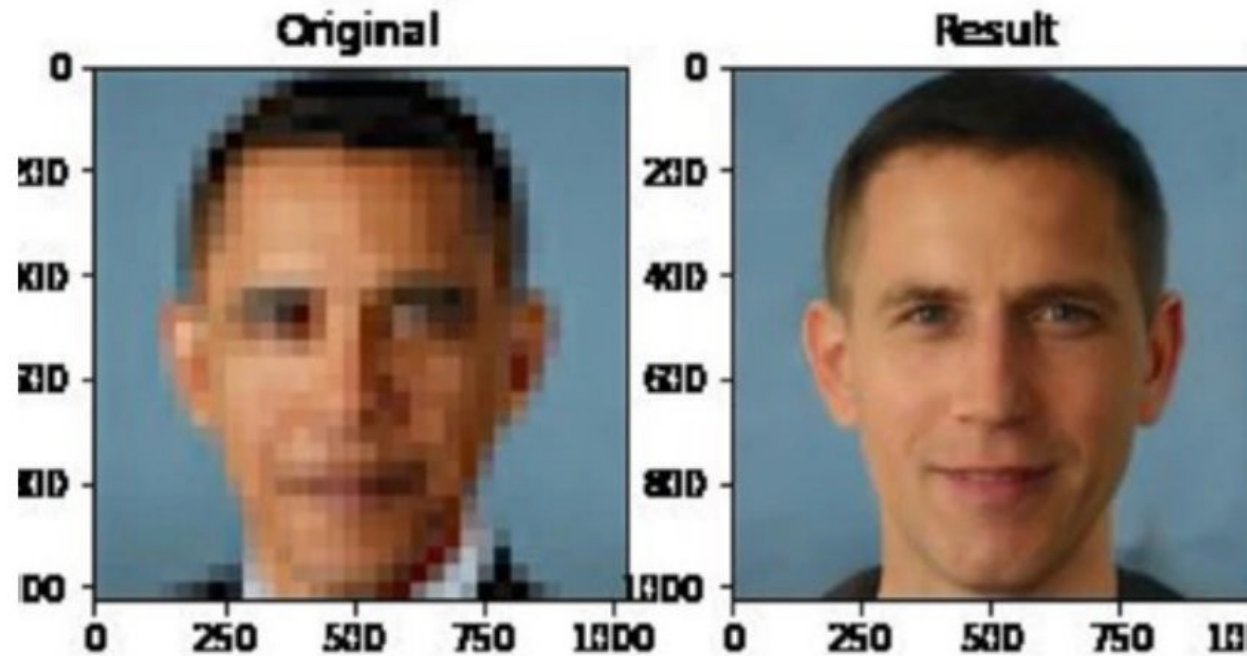
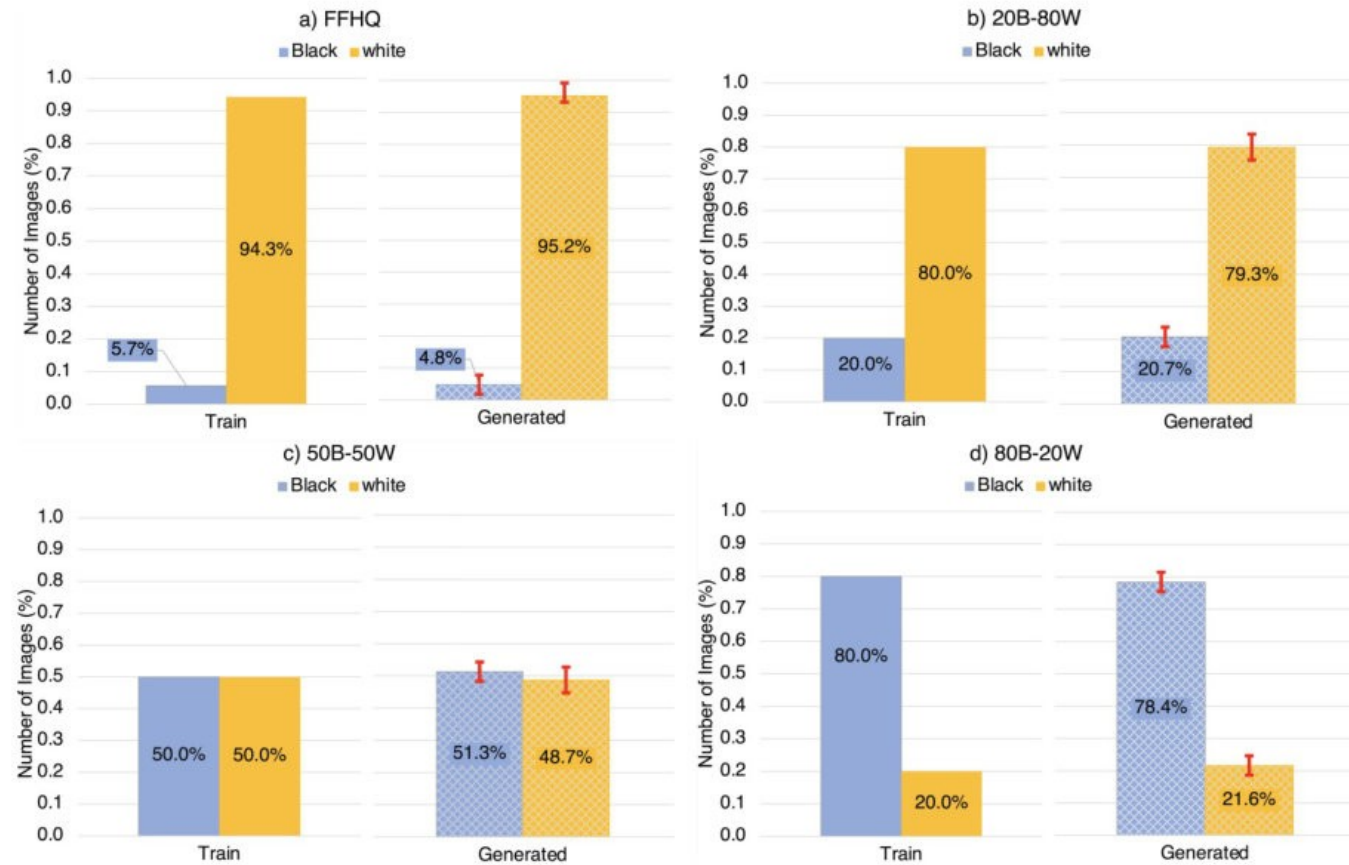


Image Super-resolution system PULSE [Menon et al., CVPR 2020]

GAN models



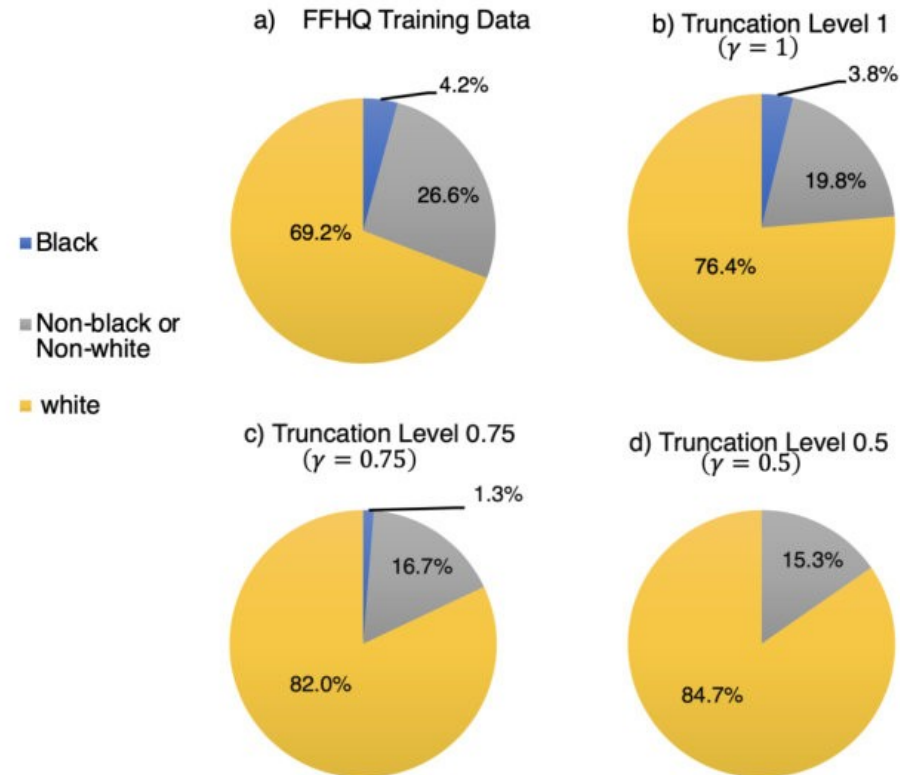
Studying Bias in GANs through the Lens of Race [Maluleke et al., 2022]

GAN models

Truncation Trick reduces diversity

$$w' = \gamma w + (1 - \gamma) \bar{w}$$

↑ ↑
Sampled code Average code



Studying Bias in GANs through the Lens of Race [Maluleke et al., 2022]

Text-to-Image Models



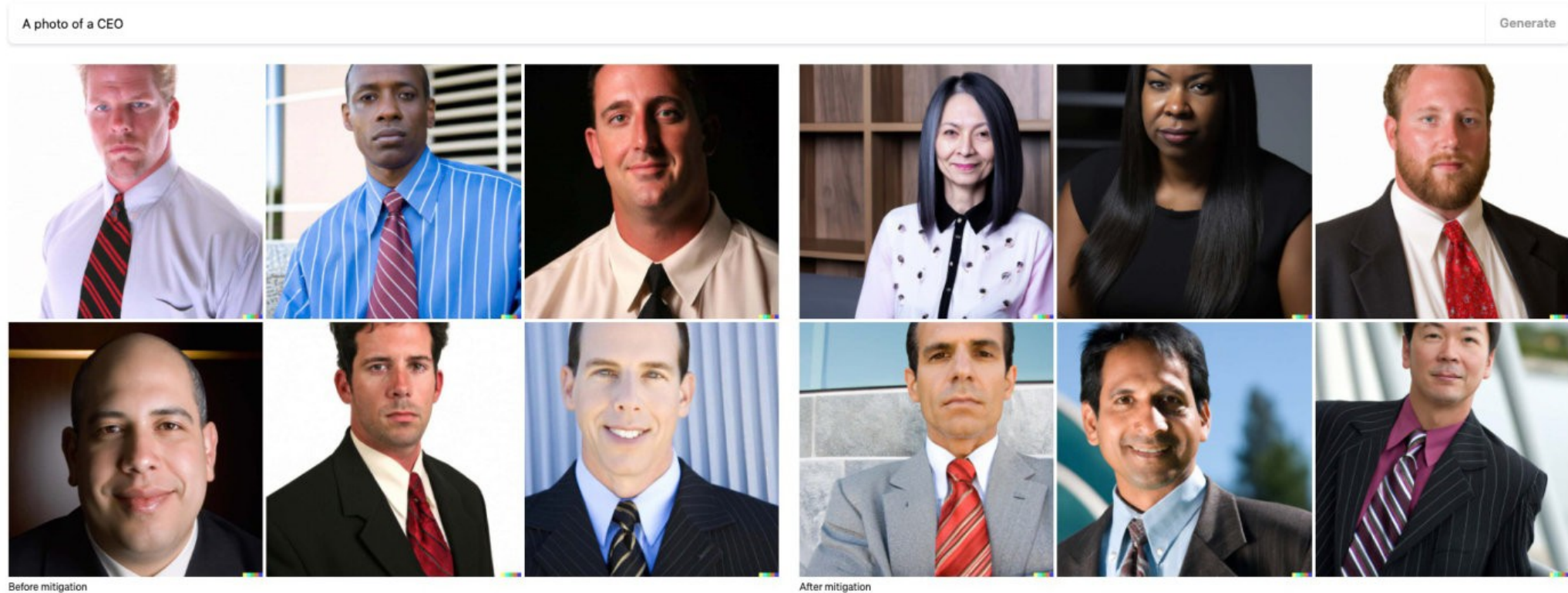
Managers



Native Americans

<https://www.technologyreview.com/2023/03/22/1070167/these-news-tool-let-you-see-for-yourself-how-biased-ai-image-models-are/>

Text-to-Image Models (quick fixes?)



<https://openai.com/blog/reducing-bias-and-improving-safety-in-dall-e-2>

Quick fixes or long-term solutions?

110

Propose a dataset of your own

1. Choose 15–20 images you made or have the right to use: drawings, photos, A1 stills.
2. Decide: are you teaching a subject (a character, an object) or a style?
3. Write one caption and a trigger word. What did you choose not to label?
4. Name one image you left out, and why.
5. Write one sentence on consent: whose work or likeness is in this set?

This proposal is the starting point for Assignment 2 (due 31 October).

Whose model is it — now?

- 1 If a LoRA is only a few megabytes on top of a model trained on billions of images, how much of the output is yours?
- 2 A model can learn Greg Rutkowski's style and can be made to forget it. Who should decide which?
- 3 Would you share a LoRA of your own style? Under what terms?

One artist, one art-and-tech paper, one technique paper

1 • ART

This artist is dominating AI-generated art. And he's not happy about it.

Melissa Heikkilä, MIT Technology Review (2022) • about Greg Rutkowski

2 • ART + TECH

Ablating Concepts in Text-to-Image Diffusion Models

Kumari, Zhang, Wang, Shechtman, Zhang & Zhu, ICCV 2023 • project page

3 • TECHNIQUE

Multi-Concept Customization of Text-to-Image Diffusion (Custom Diffusion)

Kumari, Zhang, Zhang, Shechtman & Zhu, CVPR 2023 • project page

Credits and sources

TECHNIQUE SLIDES

Adapted with thanks from Jun-Yan Zhu, CMU 16-726 Learning-Based Image Synthesis (Spring 2025), Lecture 16 "Image Editing with Optimization (part II)". Figures from Gal et al. (ICLR 2023), Ruiz et al. (CVPR 2023), Kumari et al. (CVPR 2023), Hu et al. (ICLR 2022), Han et al. (SVDiff), Ye et al. (IP-Adapter).

Efficiency: Song Han, MIT 6.5940 EfficientML.ai.

ART AND SOCIETY SLIDES

Adapted with thanks from Jun-Yan Zhu, CMU 16-726 Learning-Based Image Synthesis, Lecture 20 "Visual Forensics and Societal Impacts". Figures from Carlini et al. (2023), Kumari et al. "Ablating Concepts" (2023), Menon et al. (CVPR 2020), Maluleke et al. (2022), OpenAI, and the Content Authenticity Initiative.